



NSW Education Standards Authority

2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

**General
Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks** (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–14)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

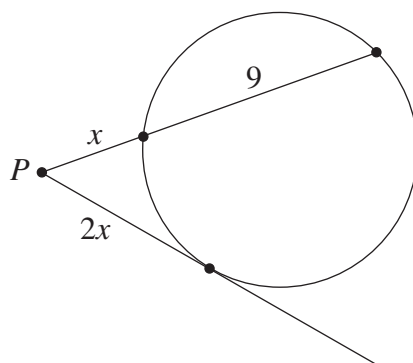
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is the domain of the function $f(x) = \ln(4 - x)$?

- A. $x < 4$
- B. $x \leq 4$
- C. $x > 4$
- D. $x \geq 4$

2 From the point P outside a circle, a secant and a tangent to the circle are constructed as shown in the diagram.



NOT TO
SCALE

Which equation is satisfied by x ?

- A. $4x^2 = 9x$
- B. $4x^2 = 9 + x$
- C. $4x^2 = 9(9 + x)$
- D. $4x^2 = x(9 + x)$

3 What is the derivative of $\tan^{-1}\frac{x}{2}$?

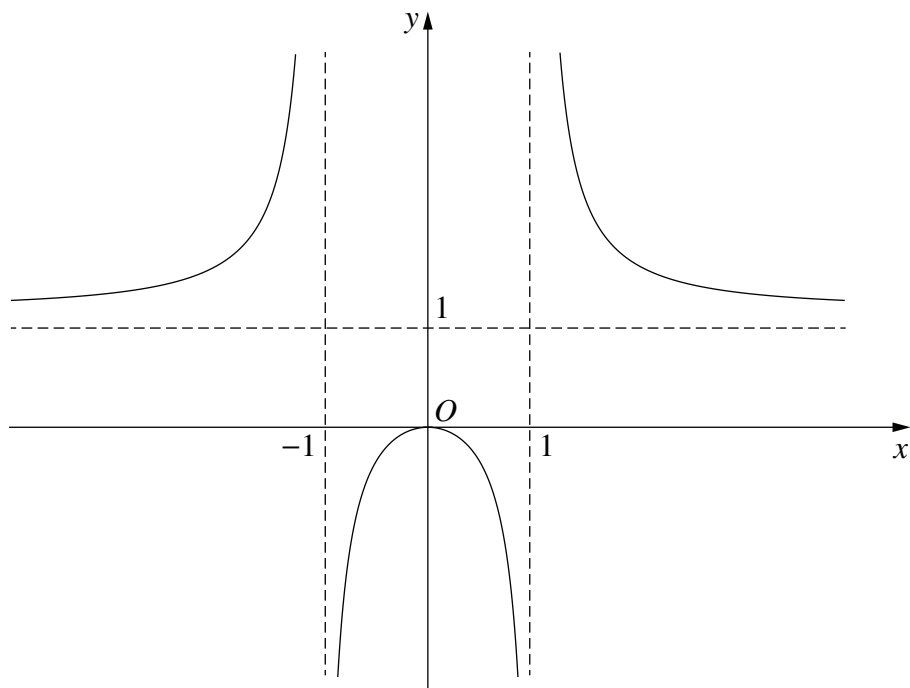
A. $\frac{1}{2(4+x^2)}$

B. $\frac{1}{4+x^2}$

C. $\frac{2}{4+x^2}$

D. $\frac{4}{4+x^2}$

4 The diagram shows the graph of $y = f(x)$.



Which equation best describes the graph?

A. $y = \frac{x}{x^2 - 1}$

B. $y = \frac{x^2}{x^2 - 1}$

C. $y = \frac{x}{1 - x^2}$

D. $y = \frac{x^2}{1 - x^2}$

- 5 A particle starts from rest, 2 metres to the right of the origin, and moves along the x -axis in simple harmonic motion with a period of 2 seconds.

Which equation could represent the motion of the particle?

- A. $x = 2 \cos \pi t$
- B. $x = 2 \cos 2t$
- C. $x = 2 + 2 \sin \pi t$
- D. $x = 2 + 2 \sin 2t$

- 6 It is given that $\sin x = \frac{1}{4}$, where $\frac{\pi}{2} < x < \pi$.

What is the value of $\sin 2x$?

- A. $-\frac{7}{8}$
- B. $-\frac{\sqrt{15}}{8}$
- C. $\frac{\sqrt{15}}{8}$
- D. $\frac{7}{8}$

- 7 Let $P(x) = qx^3 + rx^2 + rx + q$ where q and r are constants, $q \neq 0$. One of the zeros of $P(x)$ is -1 .

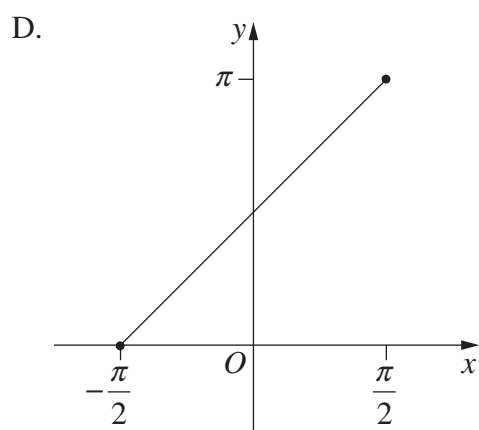
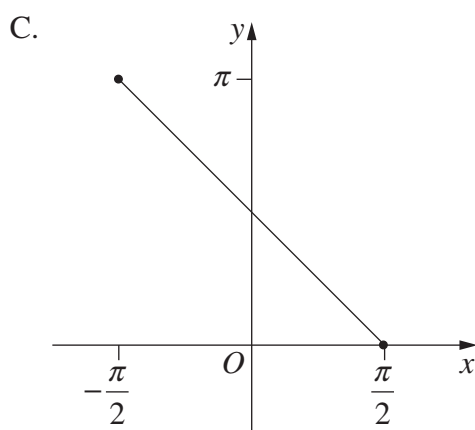
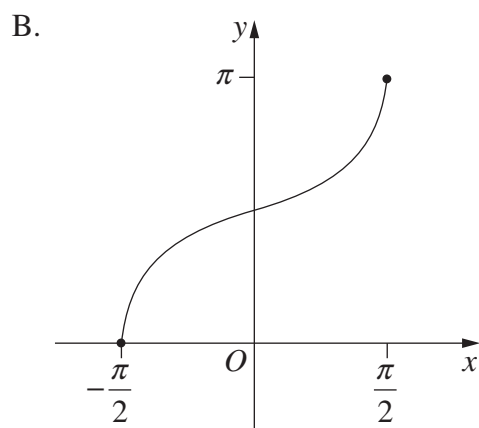
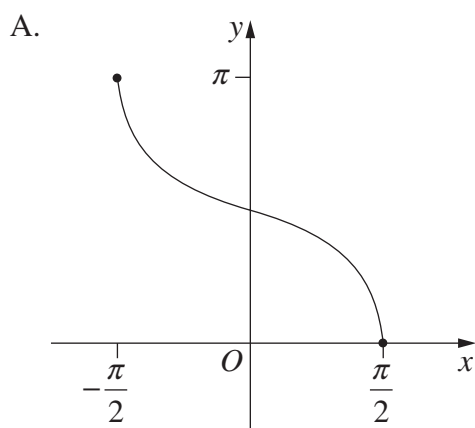
Given that α is a zero of $P(x)$, $\alpha \neq -1$, which of the following is also a zero?

- A. $-\frac{1}{\alpha}$
- B. $-\frac{q}{\alpha}$
- C. $\frac{1}{\alpha}$
- D. $\frac{q}{\alpha}$

- 8 In how many ways can all the letters of the word PARALLEL be placed in a line with the three Ls together?

- A. $\frac{6!}{2!}$
 B. $\frac{6!}{2!3!}$
 C. $\frac{8!}{2!}$
 D. $\frac{8!}{2!3!}$

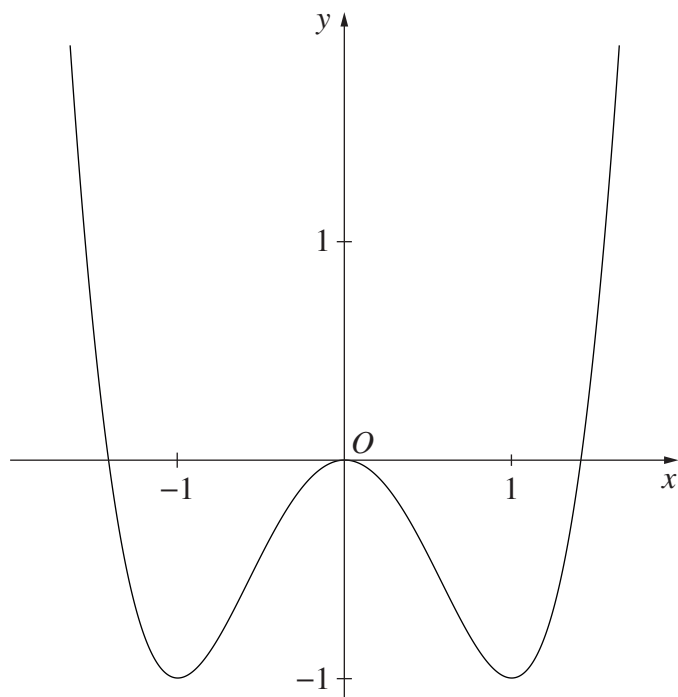
- 9 Which graph best represents $y = \cos^{-1}(-\sin x)$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$?



- 10 The function $f(x) = -\sqrt{1+\sqrt{1+x}}$ has inverse $f^{-1}(x)$.

The graph of $y = f^{-1}(x)$ forms part of the curve $y = x^4 - 2x^2$.

The diagram shows the curve $y = x^4 - 2x^2$.



How many points do the graphs of $y = f(x)$ and $y = f^{-1}(x)$ have in common?

- A. 1
- B. 2
- C. 3
- D. 4

Section II

60 marks

Attempt Questions 11–14

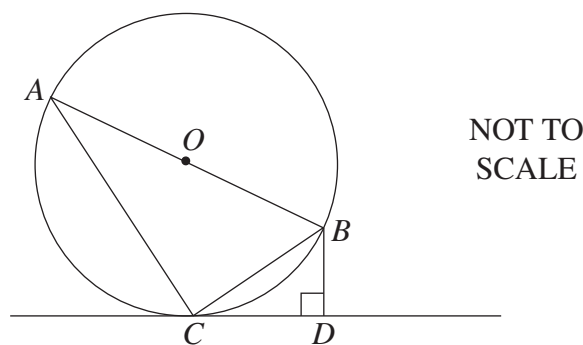
Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

- (a) Find the acute angle between the lines $x - 2y + 1 = 0$ and $y = 3x - 4$. 2
- (b) For what values of x is $\frac{x}{x+1} < 2$? 3
- (c) The line segment AB is a diameter of the circle, centred at O . The line CD is the tangent to the circle at C and BD is perpendicular to CD , as shown in the diagram. 3



Prove that $\angle ABC = \angle CBD$.

- (d) Find the polynomial $Q(x)$ that satisfies $x^3 + 2x^2 - 3x - 7 = (x - 2)Q(x) + 3$. 2

Question 11 continues on page 8

Question 11 (continued)

(e) Find $\int 2 \sin^2 4x \, dx$. **2**

(f) Prize-winning symbols are printed on 5% of ice-cream sticks. The ice-creams are randomly packed into boxes of 8.

(i) What is the probability that a box contains no prize-winning symbols? **1**

(ii) What is the probability that a box contains at least 2 prize-winning symbols? **2**

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) Distance A is inversely proportional to distance B , such that $A = \frac{9}{B}$, where A and B are measured in metres. The two distances vary with respect to time. Distance B is increasing at a rate of 0.2 m s^{-1} . **3**

What is the value of $\frac{dA}{dt}$ when $A = 12$?

- (b) A particle is moving along the x -axis in simple harmonic motion. The position of the particle is given by

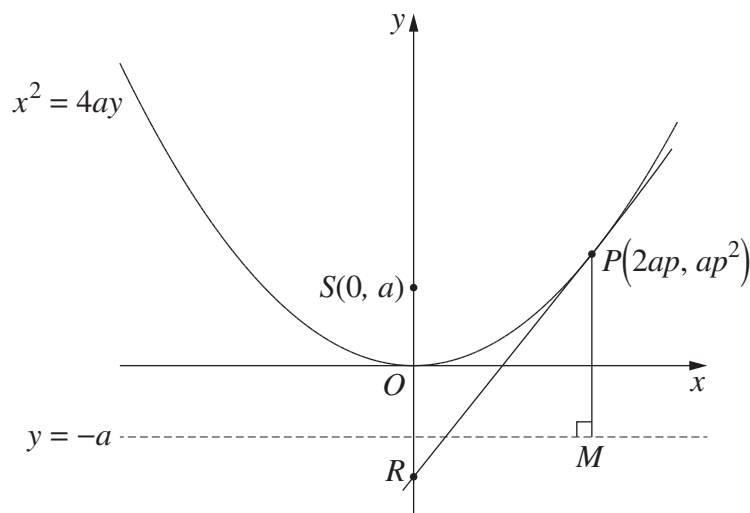
$$x = \sqrt{2} \cos 3t + \sqrt{6} \sin 3t, \text{ for } t \geq 0.$$

- (i) Write x in the form $R \cos(3t - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. **2**
- (ii) Find the two values for x where the particle comes to rest. **1**
- (iii) When is the first time that the speed of the particle is equal to half of its maximum speed? **2**

Question 12 continues on page 10

Question 12 (continued)

- (c) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$. The focus of the parabola is at $S(0, a)$ and the directrix is the line $y = -a$. The vertical line through P meets the directrix at M . The tangent at P meets the y -axis at R . **3**



Show that $\angle SPR = \angle SRP$.

- (d) A refrigerator has a constant temperature of 3°C . A can of drink with temperature 30°C is placed in the refrigerator.

After being in the refrigerator for 15 minutes, the temperature of the can of drink is 28°C .

The change in the temperature of the can of drink can be modelled by $\frac{dT}{dt} = k(T - 3)$, where T is the temperature of the can of drink, t is the time in minutes after the can is placed in the refrigerator and k is a constant.

- (i) Show that $T = 3 + Ae^{kt}$, where A is a constant, satisfies $\frac{dT}{dt} = k(T - 3)$. **1**
- (ii) After 60 minutes, at what rate is the temperature of the can of drink changing? **3**

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

(a) Use the substitution $u = \cos^2 x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin 2x}{4 + \cos^2 x} dx$. **3**

(b) In the expansion of $(5x + 2)^{20}$, the coefficients of x^k and x^{k+1} are equal. **3**

What is the value of k ?

(c) A particle moves in a straight line. At time t seconds the particle has a displacement of x m, a velocity of $v \text{ m s}^{-1}$ and acceleration $a \text{ m s}^{-2}$. Initially the particle has displacement 0 m and velocity 2 m s^{-1} . The acceleration is given by $a = -2e^{-x}$. The velocity of the particle is always positive.

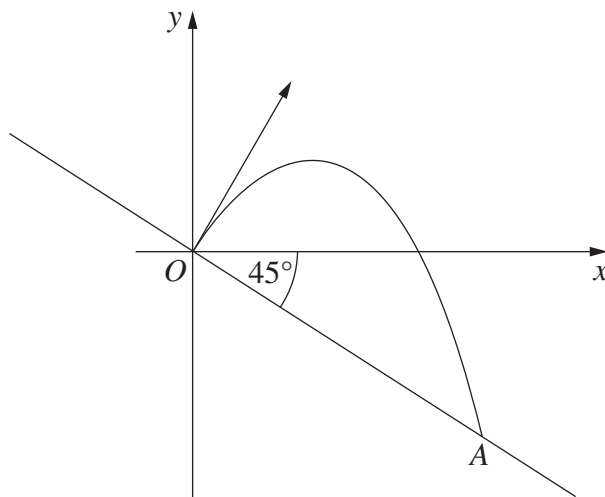
(i) Show that $v = 2e^{\frac{-x}{2}}$. **2**

(ii) Find an expression for x as a function of t . **2**

Question 13 continues on page 12

Question 13 (continued)

- (d) The point O is on a sloping plane that forms an angle of 45° to the horizontal. A particle is projected from the point O . The particle hits a point A on the sloping plane as shown in the diagram.



The equation of the line OA is $y = -x$. The equations of motion of the particle are

$$x = 18t$$

$$y = 18\sqrt{3}t - 5t^2,$$

where t is the time in seconds after projection. Do NOT prove these equations.

- | | | |
|------|---|----------|
| (i) | Find the distance OA between the point of projection and the point where the particle hits the sloping plane. | 2 |
| (ii) | What is the size of the acute angle that the path of the particle makes with the sloping plane as the particle hits the point A ? | 3 |

End of Question 13

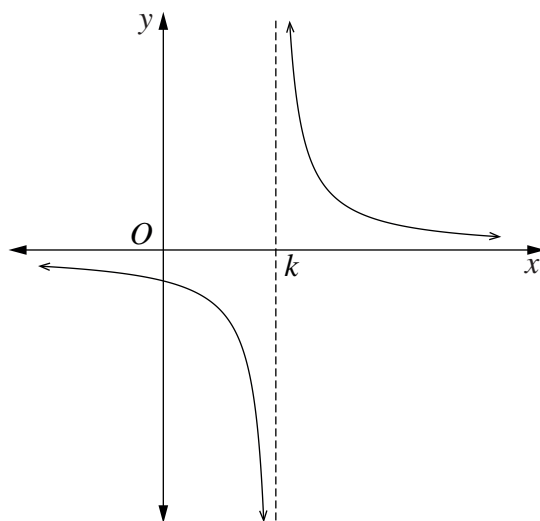
Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) Prove by mathematical induction that, for all integers $n \geq 1$,

3

$$1(1!) + 2(2!) + 3(3!) + \cdots + n(n!) = (n+1)! - 1.$$

- (b) The diagram shows the graph of $y = \frac{1}{x-k}$, where k is a positive real number.



- (i) By considering the graphs of $y = x^2$ and $y = \frac{1}{x-k}$, explain why the function $f(x) = x^3 - kx^2 - 1$ has exactly one real zero.

2

Let this zero be α and let $x_1 = k$ be a first approximation to α .

- (ii) Show that using one application of Newton's method gives a second approximation, $x_2 = k + \frac{1}{k^2}$.

2

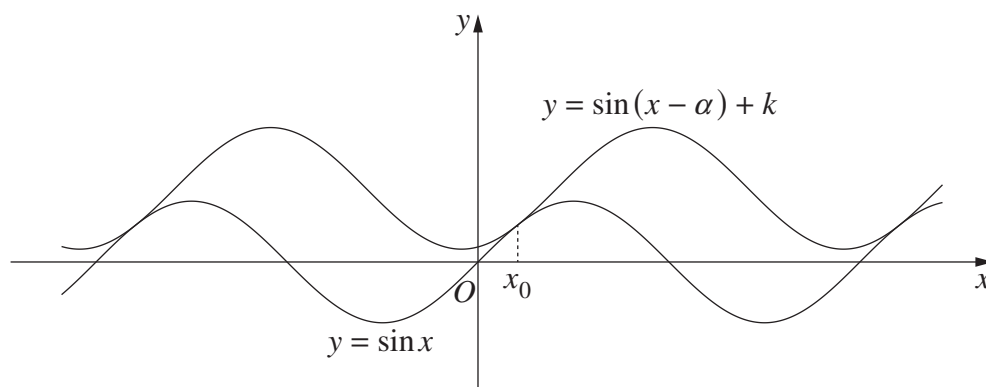
- (iii) Show that $x_1 < \alpha < x_2$.

3

Question 14 continues on page 14

Question 14 (continued)

- (c) The diagram shows the two curves $y = \sin x$ and $y = \sin(x - \alpha) + k$, where $0 < \alpha < \pi$ and $k > 0$. The two curves have a common tangent at x_0 where $0 < x_0 < \frac{\pi}{2}$.



- | | |
|--|----------|
| (i) Explain why $\cos x_0 = \cos(x_0 - \alpha)$. | 1 |
| (ii) Show that $\sin x_0 = -\sin(x_0 - \alpha)$. | 2 |
| (iii) Hence, or otherwise, find k in terms of α . | 2 |

End of paper

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2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

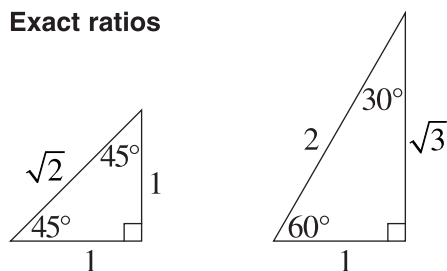
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2019 HSC Mathematics Extension 1 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	D
3	C
4	B
5	A
6	B
7	C
8	A
9	D
10	A

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Finds the slope of $x - 2y + 1 = 0$, or equivalent merit	1

Sample answer:

$$x - 2y + 1 = 0 \qquad y = 3x - 4$$

$$m_1 = \frac{1}{2} \qquad m_2 = 3$$

$$\tan \theta = \left| \frac{3 - \frac{1}{2}}{1 + 3\left(\frac{1}{2}\right)} \right|$$

$$= \frac{5}{2} \div \frac{5}{2}$$

$$\tan \theta = 1$$

$$\therefore \theta = \frac{\pi}{4}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	3
• Factorises a relevant quadratic, or equivalent merit	2
• Observes that $x \neq -1$, or equivalent merit	1

Sample answer:

$$\frac{x}{x+1} < 2$$

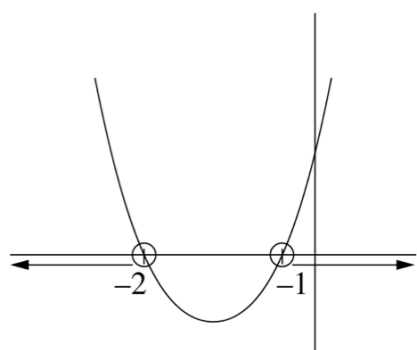
$$(x+1)^2 \left(\frac{x}{x+1} \right) < 2(x+1)^2$$

$$x(x+1) < 2(x+1)^2$$

$$0 < 2(x+1)^2 - x(x+1)$$

$$0 < (x+1)[2(x+1) - x]$$

$$0 < (x+1)(x+2)$$

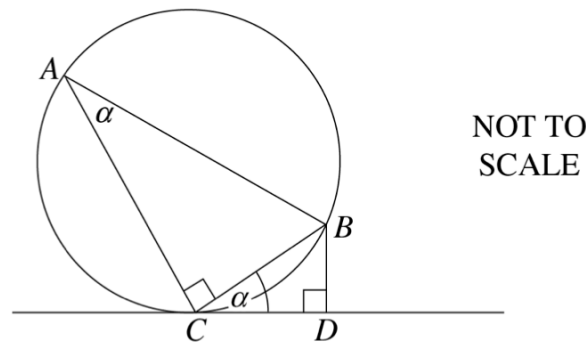


$$x < -2, x > -1$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	3
• Shows two pairs of angles are equal with correct reasons, or equivalent merit	2
• Observes that $\angle ACB$ is a right angle, or equivalent merit	1

Sample answer:



Let $\angle BCD = \alpha$
 $\therefore \angle CAB = \alpha$ (angle in alt segment)
 $\angle ACB = 90^\circ$ (angle in a semi-circle)
 $\therefore \angle ABC = 90 - \alpha$ (angle sum $\triangle$)
 $\angle CBD = 90 - \alpha$ (angle sum $\triangle$)
 $\therefore \angle ABC = \angle CBD$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Attempts long division of relevant polynomials, or equivalent merit	1

Sample answer:

$$\begin{array}{r}
 x^2 + 4x + 5 \\
 x - 2 \overline{) x^3 + 2x^2 - 3x - 7} \\
 \underline{x^3 - 2x^2} \\
 4x^2 - 3x \\
 \underline{4x^2 - 8x} \\
 5x - 7 \\
 \underline{5x - 10} \\
 3
 \end{array}$$

$$x^3 + 2x^2 - 3x - 7 = (x - 2)(x^2 + 4x + 5) + 3$$

$$\therefore Q(x) = x^2 + 4x + 5$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	2
• Correctly uses double angle result, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int 2\sin^2 4x \, dx \\
 &= \int 2 \left(\frac{1 - \cos 2(4x)}{2} \right) dx \\
 &= \int 1 - \cos 8x \, dx \\
 &= x - \frac{\sin 8x}{8} + C
 \end{aligned}$$

Question 11 (f) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$p = 0.05 \quad n = 8$$

$$q = 0.95$$

$$P(0 \text{ wins}) = (0.95)^8 \\ \doteq 0.663$$

Question 11 (f) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly finds one non-trivial probability, or equivalent merit	1

Sample answer:

$$P(\text{at least } 2) = 1 - P(0 \text{ or } 1)$$

$$= 1 - [0.95^8 + {}^8C_1 (0.05)^1 (0.95)^7]$$

$$\doteq 0.057$$

Question 12 (a)

Criteria	Marks
• Provides correct solution	3
• Finds $\frac{dA}{dt}$ in terms of B , or equivalent merit	2
• Finds $\frac{dA}{dB}$, or equivalent merit	1

Sample answer:

$$A = 9B^{-1}$$

$$\frac{dA}{dB} = -9B^{-2}$$

$$= \frac{-9}{B^2}$$

$$A = 12$$

$$12 = \frac{9}{B}$$

$$B = 0.75$$

$$\frac{dB}{dt} = 0.2$$

$$\begin{aligned} \therefore \frac{dA}{dt} &= \frac{dA}{dB} \times \frac{dB}{dt} \\ &= \frac{-9}{(0.75)^2} \times 0.2 \end{aligned}$$

$$\therefore \frac{dA}{dt} = -3.2$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Finds R , or equivalent merit	1

Sample answer:

$$R\cos(3t - \alpha) = R\cos 3t \cos \alpha + R\sin 3t \sin \alpha$$

$$\text{Let } R\cos 3t \cos \alpha + R\sin 3t \sin \alpha = \sqrt{2} \cos 3t + \sqrt{6} \sin 3t$$

$$\therefore R\cos \alpha = \sqrt{2} \quad \text{--- ①} \quad R\sin \alpha = \sqrt{6} \quad \text{--- ②}$$

$$\text{①}^2 + \text{②}^2 \quad R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = (\sqrt{2})^2 + (\sqrt{6})^2$$

$$R^2 = 2 + 6$$

$$R = 2\sqrt{2}$$

$$\frac{\text{②}}{\text{①}} \quad \frac{R\sin \alpha}{R\cos \alpha} = \frac{\sqrt{6}}{\sqrt{2}}$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore x = 2\sqrt{2} \cos\left(3t - \frac{\pi}{3}\right)$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct answers	1

Sample answer:

At rest when $x = \pm 2\sqrt{2}$.

Question 12 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds half the maximum speed, or equivalent merit	1

Sample answer:

$$x = 2\sqrt{2} \cos\left(3t - \frac{\pi}{3}\right)$$

$$v = -6\sqrt{2} \sin\left(3t - \frac{\pi}{3}\right)$$

$$\text{Max speed} = +6\sqrt{2}$$

$$\text{Half max speed} = 3\sqrt{2}$$

$$\therefore -\frac{1}{2} = \sin\left(3t - \frac{\pi}{3}\right)$$

$$3t - \frac{\pi}{3} = -\frac{\pi}{6}$$

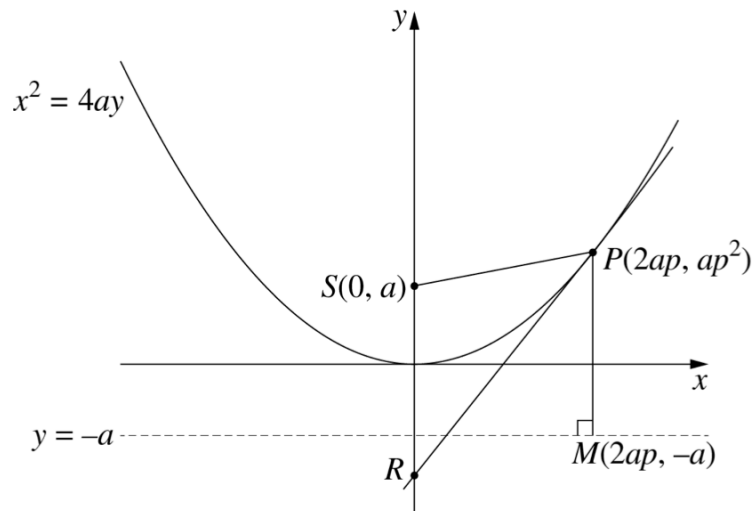
$$3t = \frac{\pi}{6}$$

$$t = \frac{\pi}{18}$$

Question 12 (c)

Criteria	Marks
• Provides correct solution	3
• Finds the length of SP , or equivalent merit	2
• Finds coordinates of R , or equivalent merit	1

Sample answer:



$$SP = PM \qquad PM = ap^2 + a$$

Point P is equidistant to $S(0, a)$ and directrix.

$$\therefore SP = ap^2 + a \quad \text{--- ①}$$

$$\text{tangent at } P \rightarrow y = px - ap^2$$

$$\text{at } R \quad x = 0$$

$$\therefore y = -ap^2 \text{ ie } R(0, -ap^2)$$

$$\therefore SR = a + ap^2$$

$$\therefore SR = SP \quad \text{from ①}$$

$$\therefore \text{In } \triangle SPR$$

$$SP = SR$$

$$\therefore \angle SPR = \angle SRP \quad (\text{base } \angle \text{s of isosceles})$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$T = 3 + Ae^{kt} \quad Ae^{kt} = T - 3$$

$$\frac{dT}{dt} = k.Ae^{kt}$$

$$\therefore \frac{dT}{dt} = k(T - 3)$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds the value of k , or equivalent merit	2
• Finds the value of A , or equivalent merit	1

Sample answer:

$$t = 60$$

$$\frac{dT}{dt} = ?$$

$$t = 0 \quad T = 30 \quad T = 3 + Ae^{kt}$$

$$30 = 3 + Ae^0$$

$$A = 27$$

$$t = 15 \quad T = 28 \quad 28 = 3 + 27e^{15k}$$

$$\frac{25}{27} = e^{15k}$$

$$k = \frac{1}{15} \ln\left(\frac{25}{27}\right)$$

$$t = 60 \quad T = 3 + 27e^{k \cdot 60} \\ = 22.8458...$$

$$\therefore t = 60 \quad \frac{dT}{dt} = \frac{1}{15} \ln\left(\frac{25}{27}\right)(22.8458 - 3) \\ = -0.1018...$$

$\therefore$ Temperature is decreasing at 0.102° per min.

Question 13 (a)

Criteria	Marks
• Provides correct solution	3
• Finds correct primitive, or equivalent merit	2
• Finds $\frac{du}{dx}$, or equivalent merit	1

Sample answer:

$$\int_0^{\frac{\pi}{4}} \frac{\sin 2x}{4 + \cos^2 x} dx$$

$$\begin{aligned} \text{let } u &= \cos^2 x \\ \frac{du}{dx} &= -2\cos x \sin x \\ \therefore du &= -\sin 2x dx \end{aligned}$$

$$\text{when } x = 0 \quad u = 1$$

$$\text{when } x = \frac{\pi}{4} \quad u = \frac{1}{2}$$

$$\begin{aligned} \therefore \text{The integral} &= \int_1^{\frac{1}{2}} \frac{-du}{4+u} \\ &= \int_{\frac{1}{2}}^1 \frac{du}{4+u} \\ &= \left[\ln(4+u) \right]_{\frac{1}{2}}^1 \\ &= \ln 5 - \ln \frac{9}{2} \\ &= \ln \left(\frac{10}{9} \right) \end{aligned}$$

Question 13 (b)

Criteria	Marks
• Provides correct solution	3
• Finds a correct equation in k , or equivalent merit	2
• Uses the binomial theorem to find a coefficient, or equivalent merit	1

Sample answer:

$$\text{Coefficient of } x^k = \binom{20}{k} 5^k 2^{20-k}$$

$$\text{Coefficient of } x^{k+1} = \binom{20}{k+1} 5^{k+1} 2^{19-k}$$

Equating these coefficients,

$$\frac{20!}{k!(20-k)!} 5^k 2^{20-k} = \frac{20!}{(k+1)!(19-k)!} 5^{k+1} 2^{19-k}$$

$$(k+1)2 = (20-k)5$$

$$2k + 2 = 100 - 5k$$

$$7k = 98$$

$$k = 14$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Integrates to find $\frac{1}{2}v^2$, or equivalent merit	1

Sample answer:

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -2e^{-x}$$

$$\frac{1}{2}v^2 = \int -2e^{-x} dx$$

$$= 2e^{-x} + c$$

when $x = 0$ $v = 2$

$$\frac{1}{2} \times 2^2 = 2e^0 + c$$

$$\therefore c = 0$$

$$\therefore v^2 = 4e^{-x}$$

$$\therefore v = \pm 2e^{-\frac{x}{2}}$$

We disregard the negative case since it is given that v is always positive

$$\therefore v = 2e^{-\frac{x}{2}}$$

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Integrates to find t , or equivalent merit	1

Sample answer:

$$\frac{dx}{dt} = 2e^{-\frac{x}{2}}$$

$$\frac{dt}{dx} = \frac{1}{2}e^{\frac{x}{2}}$$

$$t = \int \frac{1}{2}e^{\frac{x}{2}} dx$$

$$= e^{\frac{x}{2}} + C_1$$

when $t = 0$ $x = 0$

$$0 = e^0 + C_1$$

$$\therefore C_1 = -1$$

$$t = e^{\frac{x}{2}} - 1$$

$$e^{\frac{x}{2}} = t + 1$$

$$x = 2\ln(t + 1)$$

Question 13 (d) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the time when the particle lands on the sloping plane, or equivalent merit	1

Sample answer:

We want $y = -x$

$$18\sqrt{3}t - 5t^2 = -18t$$

$$\therefore 5t^2 = (18\sqrt{3} + 18)t$$

$$\therefore t = 0 \quad \text{or} \quad t = \frac{18(\sqrt{3} + 1)}{5}$$

$$\text{At this time } x = 18 \cdot \frac{18(\sqrt{3} + 1)}{5}$$

$$\therefore OA = \sqrt{x^2 + x^2} = \sqrt{2}x$$

$$= \frac{18^2 \sqrt{2} (\sqrt{3} + 1)}{5}$$

$$\doteq 250.37 \text{ m}$$

Question 13 (d) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds the slope of the path at A, or equivalent merit	2
• Finds a relevant derivative, or equivalent merit	1

Sample answer:

Find the components of the velocity

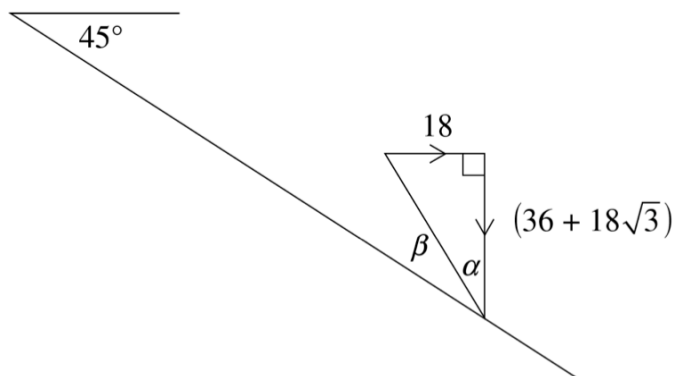
$$\dot{x} = 18$$

$$\dot{y} = 18\sqrt{3} - 10t$$

when $t = \frac{18(\sqrt{3} + 1)}{5}$

$$\dot{y} = 18\sqrt{3} - (36\sqrt{3} + 36)$$

$$= -36 - 18\sqrt{3}$$



$$\tan \alpha = \frac{18}{36 + 18\sqrt{3}}$$

$$= \frac{1}{2 + \sqrt{3}}$$

$$\div 0.2679$$

$$\therefore \alpha = 15^\circ$$

$\therefore$ angle with sloping plane is $\beta = 45^\circ - \alpha = 30^\circ$.

Question 14 (a)

Criteria	Marks
• Provides correct solution	3
• Establishes the inductive step, or equivalent merit	2
• Establishes the $n = 1$ case, or equivalent merit	1

Sample answer:

$$n = 1: \quad \text{LHS} = 1(1!) = 1$$

$$\text{RHS} = (1 + 1)! - 1 = 1 = \text{LHS}$$

True when $n = 1$

Suppose true for $n = k$

$$\text{So } 1(1!) + 2(2!) + \cdots + k(k!) = (k + 1)! - 1$$

Want to show it is then true for $n = k + 1$

$$\text{that is } 1(1!) + \cdots + k(k!) + (k + 1)((k + 1)!) = ((k + 1) + 1)! - 1$$

$$\begin{aligned} \text{LHS} &= 1(1!) + \cdots + k(k!) + (k + 1)((k + 1)!) \\ &= ((k + 1)! - 1) + (k + 1)((k + 1)!) \\ &= (k + 1)! + (k + 1)(k + 1)! - 1 \\ &= (k + 1)!(1 + k + 1) - 1 \\ &= (k + 1)!(k + 2) - 1 \\ &= (k + 2)! - 1 \\ &= \text{RHS} \end{aligned}$$

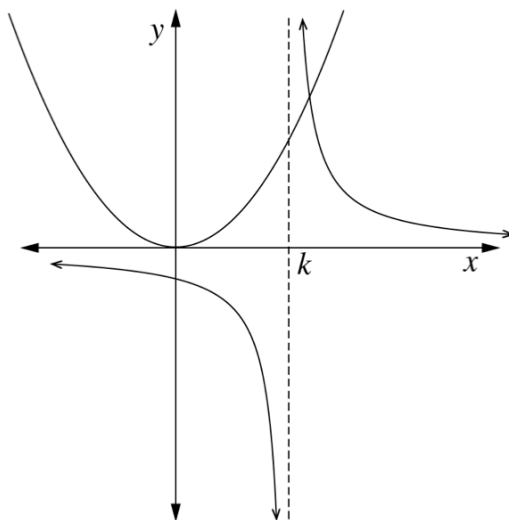
If true for $n = k$, then true for $n = k + 1$

Hence, by mathematical induction, true for $n \geq 1$.

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	2
• States that the two curves intersect at exactly one point, or equivalent merit	1

Sample answer:



Adding $y = x^2$ to the sketch we can see that there is exactly one point of intersection, so $x^2 = \frac{1}{x-k}$ has exactly one solution.

If $f(x) = 0$

then $x^3 - kx^2 - 1 = 0$

$$x^2(x-k) = 1$$

$$x^2 = \frac{1}{x-k}$$

which has exactly one solution.

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses Newton's method, or equivalent merit	1

Sample answer:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f'(x) = 3x^2 - 2kx$$

$$\begin{aligned} \text{so } x_2 &= k - \frac{(k^3 - k^3 - 1)}{3k^2 - 2k^2} \\ &= k + \frac{1}{k^2} \end{aligned}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Shows $f(x_2) > 0$, or equivalent merit	2
• Shows $f(x_1) < 0$, or equivalent merit	1

Sample answer:

$$f(x_1) = f(k)$$

$$= k^3 - k^3 - 1$$

$$= -1$$

$$< 0$$

$$f(x_2) = \left(k + \frac{1}{k^2}\right)^3 - k\left(k + \frac{1}{k^2}\right)^2 - 1$$

$$= k^3 + 3 + \frac{3}{k^3} + \frac{1}{k^6} - k^3 - 2 - \frac{1}{k^3} - 1$$

$$= \frac{2}{k^3} + \frac{1}{k^6}$$

$$> 0$$

as k is positive.

As $f(x_1) < 0$ and $f(x_2) > 0$ there must be a zero of $f(x)$ between x_1 and x_2 .

As α is the only zero of $f(x)$ we have $x_1 < \alpha < x_2$.

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

The curves have a common tangent at x_0 and so have the same slope there.

$$y = \sin x$$

$$\frac{dy}{dx} = \cos x$$

$$y = \sin(x - \alpha) + k$$

$$\frac{dy}{dx} = \cos(x - \alpha)$$

and so $\cos x_0 = \cos(x_0 - \alpha)$.

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Observes that $\sin x_0 = \sin(x_0 - \alpha) + k$, or equivalent merit	1

Sample answer:

As $\cos x_0 = \cos(x_0 - \alpha)$

We have

$$\begin{aligned} \sin^2 x_0 &= 1 - \cos^2 x_0 \\ &= 1 - \cos^2(x_0 - \alpha) \quad (\text{from (i)}) \end{aligned}$$

$$= \sin^2(x_0 - \alpha)$$

$$\sin x_0 = \pm \sin(x_0 - \alpha)$$

But at x_0

$$\sin x_0 = \sin(x_0 - \alpha) + k$$

with $k > 0$

so $\sin x_0 \neq \sin(x_0 - \alpha)$

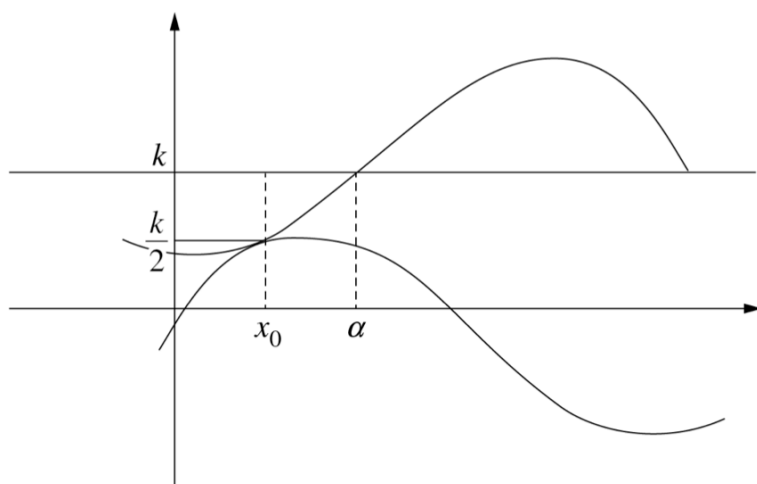
Hence $\sin x_0 = -\sin(x_0 - \alpha)$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Shows $\sin x_0 = \frac{k}{2}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \text{At } x_0 \quad \sin x_0 &= \sin(x_0 - \alpha) + k \\
 \sin x_0 &= -\sin x_0 + k && \text{from (ii)} \\
 2\sin x_0 &= k \\
 \sin x_0 &= \frac{k}{2}
 \end{aligned}$$



Now $0 < \alpha < \pi$ and $0 < x_0 < \frac{\pi}{2}$

So $-\pi < x_0 - \alpha < \frac{\pi}{2}$

$$\cos(x_0 - \alpha) = \cos x_0$$

$$\sin(x_0 - \alpha) = -\sin x_0$$

Hence $x_0 - \alpha = 2n\pi - x_0$

and so $x_0 - \alpha = -x_0$ as $-\pi < x_0 - \alpha < \frac{\pi}{2}$

$$2x_0 = \alpha$$

$$x_0 = \frac{\alpha}{2}$$

Thus $k = 2\sin \frac{\alpha}{2}$

2019 HSC Mathematics Extension 1

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	4.1, 12.2	HE7
2	1	2.9	PE3
3	1	15.5	HE4
4	1	10.5	HE4
5	1	14.4	HE3
6	1	5.7	PE6
7	1	16.3	PE3
8	1	18.1	PE3
9	1	15.3	HE4
10	1	15.1	HE4

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	6.6	PE6
11 (b)	3	1.4	PE3
11 (c)	3	2.9	PE3
11 (d)	2	16.2	PE3
11 (e)	2	13.6	HE6
11 (f) (i)	1	18.2	HE3
11 (f) (ii)	2	18.2	HE3
12 (a)	3	14.1	HE5
12 (b) (i)	2	5.9	HE3
12 (b) (ii)	1	14.4	HE3
12 (b) (iii)	2	14.4	HE3
12 (c)	3	9.6	PE3, PE4
12 (d) (i)	1	14.2	HE3
12 (d) (ii)	3	14.2	HE3
13 (a)	3	5.9, 11.5	HE6
13 (b)	3	17.3	HE2
13 (c) (i)	2	14.3	HE5
13 (c) (ii)	2	14.3	HE5
13 (d) (i)	2	14.3	HE3
13 (d) (ii)	3	14.3	HE3
14 (a)	3	7.4	HE2
14 (b) (i)	2	4.3	PE3, PE6
14 (b) (ii)	2	16.4	PE3, HE4

Question	Marks	Content	Syllabus outcomes
14 (b) (iii)	3	16.1, 16.4	HE7
14 (c) (i)	1	13.5, 13.7	HE7
14 (c) (ii)	2	13.5, 13.7	HE7
14 (c) (iii)	2	13.5, 13.7	HE7